

Question Number : 181 Question Id : 640653739753 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Correct Marks : 1

Question Label : Short Answer Question

Enter the correct answer for blank (b) _____.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

LLM

Section Id :	64065351461
Section Number :	12
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	12
Number of Questions to be attempted :	12
Section Marks :	50
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1

Sub-Section Id :

640653108276

Question Shuffling Allowed :

No

Is Section Default? :

null

Question Number : 182

Question Id : 640653739754

Question Type : MCQ

Is Question Mandatory : No

Calculator : None

Response Time : N.A

Think Time : N.A

Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INTRODUCTION TO LARGE LANGUAGE MODELS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532474501.  YES

6406532474502.  NO

Question Number : 183

Question Id : 640653739755

Question Type : MCQ

Is Question Mandatory : No

Calculator : None

Response Time : N.A

Think Time : N.A

Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

General Note:

- If you find the given information is insufficient in any of the Numerical Answer Type (NAT) questions, enter -1 as the answer.
- In all your calculations, take at least two digits. For example, if the intermediate result is 1.860875, then take it as 1.86 to the next step.

Options :

6406532474503. ✔ Useful Data has been mentioned above.

6406532474504. ✖ This data attachment is just for a reference & not for an evaluation.

Sub-Section Number :	2
Sub-Section Id :	640653108277
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653739756 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (184 to 187)

Question Label : Comprehension

Consider a vocabulary $\mathcal{V} = (A, T, C, G)$ and the corresponding

embedding matrix $E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}$. For example, the 0-th

row of E is the embedding for 0-th token of $\mathcal{V}$ and so on.

The following sequence $X = (G, C, C, T, A, G)$ is to be processed by the encoder layer of the transformer architecture by appropriately adding the positional information using the following encoding scheme

$$P(pos, 2i) = \sin\left(\frac{pos}{10^{(2i/d_{model})}}\right)$$

$$P(pos, 2i + 1) = \cos\left(\frac{pos}{10^{(2i/d_{model})}}\right)$$

$$(A) \rightarrow [1 \ 0]$$

$$(T) \rightarrow [0 \ 1]$$

$$(C) \rightarrow [1 \ -1]$$

$$(G) \rightarrow [1 \ 1]$$

Assume the index for the position starts from zero. Add positional information to all the tokens in the sequence. Let the resultant sequence be denoted by $H = E(X) + P$, (H will be used in the subsequent questions, please write it down). Here, $E(X)$ denotes the embedding for each token in X

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 184 Question Id : 640653739757 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What is the sum of the elements in the embedding vector P_5 ?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

$$P_5 = (\sin(5), \cos(5))$$

$$= -0.95 + \downarrow$$

$$= -0.67$$

-0.73 to -0.63



Question Number : 185 Question Id : 640653739758 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

What is the sum of the elements in the 3-rd token, $h_3 \in H$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.13 to 0.20

Handwritten calculations for Question 185:

$$H = x \cdot P \quad \text{3rd}$$
$$\begin{bmatrix} 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0.14 \\ 0.02 \end{bmatrix} = 0.14 + 0.02 = 0.16$$

Matrix multiplication shown for the 3rd token (row vector [0, 1]) and the column vector [0.14, 0.02].

Question Number : 186 Question Id : 640653739759 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

The bottommost encoder layer of the transformer contains the following parameter matrices

$$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad W_K = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}, \quad W_V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad W_O = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

given that the embeddings are row vectors, compute the attention score to produce the new representation for the token (T) at the 3-rd position in the sequence X.

Handwritten calculations for Question 186:

$$Q = H_3 \times W_Q$$
$$K = H_4 \times W_K$$
$$\Phi = Q \cdot K^T$$

Diagram showing the sequence of operations: $H_3 \rightarrow Q$, $H_4 \rightarrow K$, and $Q \rightarrow \Phi$.

What is the un-normalized attention weight given for the token (A) at the 4th position?

Note: unnormalized here implies the raw score before applying softmax. Avoid scaling the dot-product by $\sqrt{d_k}$

$$\Phi_{1K}^T$$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

— 0.058

Text Areas : PlainText

Possible Answers :

-0.06 to -0.035

Question Number : 187 **Question Id :** 640653739760 **Question Type :** MCQ **Is Question**

Mandatory : No **Calculator :** None **Response Time :** N.A **Think Time :** N.A **Minimum Instruction Time :** 0

Correct Marks : 2

Question Label : Multiple Choice Question

The bottommost encoder layer of the transformer contains the following parameter matrices

$$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad W_K = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad W_V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad W_O = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

given that the embeddings are row vectors, compute the attention score to produce the new representation for the token (T) at the 3-rd position in the sequence X .

Which token in the sequence was given the highest score?

Options :

6406532474508. ✓ G



6406532474509. ✗ C

6406532474510. ✗ T

6406532474511. ✗ A

Sub-Section Number :

3

Sub-Section Id :

640653108278

Question Shuffling Allowed :

No

Is Section Default? : null

Question Id : 640653739761 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0
Question Numbers : (188 to 189)

Question Label : Comprehension

Consider the following configuration for the Vannila transformer architecture with one encoder layer and one decoder layer.

- Source and target vocabulary size = 100
- maximum sequence length = 32
- length of context window (T) for both encoder and decoder = 32
- number of heads $n_h = 2$
- $df = 4 * d_{model}$
- $dq = dk = dv = \frac{d_{model}}{n_h}$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 188 Question Id : 640653739762 Question Type : SA Calculator : None
Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Suppose the number of learnable parameters in the source input embedding layer is 1600, how many parameters are there in the positional embedding layer of the source language? Assume the positional embeddings are learnable.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

$$\frac{1600}{d_m} = \text{source } d_m$$
$$d_m = \frac{1600}{16} = 16$$

$$T \times d_m$$
$$32 \times 16$$

$$\begin{array}{r} 32 \\ 16 \\ \hline 192 \\ 32 \times \\ \hline 512 \end{array}$$

Possible Answers :

512

Question Number : 189 Question Id : 640653739763 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

How many parameters are there in the multi-head attention layer of the encoder (exclude the parameters in the W_O matrix used for linear transformation)?

Response Type : Numeric

$$MHA = 3 \times 4 \times 4 \times 16$$

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

768

Sub-Section Number :

4

Sub-Section Id :

640653108279

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 190 Question Id : 640653739764 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $R_E \in \mathbb{R}^{T_1 \times d_{model}}$ be the representation from the topmost encoder layer of the encoder and $Z_D \in \mathbb{R}^{T_2 \times d_{model}}$ be the output of Multi-head Masked Attention layer of one of the decoder layers. What will be the size of the attention matrix in the Multi-head Cross Attention layer of the same decoder layer? Assume $d_q = d_v = d_k = d$ if required.

$$\rightarrow T_2 \times T_1$$

Options :

6406532474514. ✖ $T_1 \times T_1$

6406532474515. ✖ $T_1 \times T_2$

6406532474516. ✖ $T_2 \times T_2$

6406532474517. ✔ $T_2 \times T_1$

Question Number : 191 Question Id : 640653739770 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose we run the pre-trained GPT model in an autoregressive fashion for generating a text sequence. Then, the statement that "We do not need to add the positional embedding to the predicted tokens at every time step" is

Options:

6406532474530. ✔ TRUE

6406532474531. ✖ FALSE

Sub-Section Number :	5
Sub-Section Id :	640653108280
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 192 Question Id : 640653739765 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose we have a dataset for machine translation tasks with thousands of samples. Suppose a team considers training the transformer model. The model could be trained in two approaches

A Autoregressive training ✓

B Teacher forcing ✓

Choose the correct statements

Options :

6406532474518. ✖ Approach A helps the model to converge faster than approach B

6406532474519. ✓ Approach B helps the model converge faster than approach A

6406532474520. ✓ One can start the training with approach B first and then switch to approach A after some training steps

6406532474521. ✖ Once the training starts with approach B and then switching to approach A after some training steps can not be done

Sub-Section Number :

6

Sub-Section Id :

640653108281

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 193 Question Id : 640653739766 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A team trained a Language model using the GPT architecture using a large corpus of text. The model configurations are given below

- Vocabulary size = 30000 ✓
- $d_{model} = 512$ ✓
- length of context window $T = 1024$ ✓
- $df f = 4 * d_{model}$ ✓
- $d_q = d_k = d_v = 64$ ✓

$$\frac{d_{model}}{4} = \frac{512}{4} = 128$$

$$\frac{128}{8} = 16$$

d_{model}

They trained the model for millions of steps and stopped the training after the loss had decreased significantly. Then, they used the model to generate the text by giving the special starting token [BOS] as an input to the model. Select the length of the text sequence that the model could generate.

Options :

- 6406532474522. ✓ 512
- 6406532474523. ✓ 1024
- 6406532474524. ✓ 1536
- 6406532474525. ✓ 2048

infinite pass;

Question Number : 194 Question Id : 640653739771 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Choose the correct statements

Options :

- 6406532474532. ✓ All the parameters of the GPT model are randomly initialized during pre-training
- 6406532474533. ✗ The model minimizes the CLM objective during fine-tuning to improve the performance

6406532474534. ✖ All parameters of the GPT model are randomly initialized during fine-tuning

6406532474535. ✔ In general, fine-tuning requires a dataset with labels

Question Number : 195 Question Id : 640653739772 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following decoding strategies is (are) appropriate for machine translation tasks? Assume we have infinite compute and the values for $k \geq 3, p > 0.35$ where required.

Options :

6406532474536. ✖ Top-k sampling

6406532474537. ✔ Beam search with the beam size $K = 2$

6406532474538. ✖ Top-p sampling

6406532474539. ✔ Exhaustive search

6406532474540. ✖ None of these

Sub-Section Number : 7

Sub-Section Id : 640653108282

Question Shuffling Allowed : No

Is Section Default? : null

Question Id : 640653739767 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (196 to 197)

Question Label : Comprehension

Consider a GPT model used for Causal language modelling. We feed the input sentence "This is a

cool idea” to the model by appending special staring [BOS] and ending [EOS] tokens (that is, “[BOS] This is a cool idea [EOS]”). Assume the context length of the model is 7. The attention matrix computed in one of the attention layers is given below

	Bos	.	.	.	.	Eos
Bos	1	0	0	0	0	0
.	0.5	0.5	0	0	0	0
.	0.3	0.2	0.5	0	0	0
.	0.1	0.1	0.8	0	0	0
.	0	0.15	0	0.6	0.25	0
Eos	0.25	0.35	0	0.2	0.1	0.1
	0.1	0.1	0.1	0.1	0.1	0.4

Table 1: Attention score

My
This is a
cool idea

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 196 Question Id : 640653739768 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Correct Marks : 4

Question Label : Multiple Choice Question

The attention score matrix given in Table 1 is appropriate for the causal language modelling task

Options :

- 6406532474526. ✓ True
- 6406532474527. ✗ False
- 6406532474528. ✗ Insufficient information

Question Number : 197 Question Id : 640653739769 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0 Correct Marks : 4

Question Label : Short Answer Question

Assume the time step starts from $t = 0$ and ends at $t = 6$. Suppose the model is at time step $t = 4$, what is the attention value assigned for the word "is"?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0



Sub-Section Number :

8

Sub-Section Id :

640653108283

Question Shuffling Allowed :

Yes

Is Section Default? :

null

Question Number : 198 Question Id : 640653739773 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

The statement that "the Next Sentence Prediction (NSP) task requires the BERT model to run autoregressively given the first sentence(or segment)" is

Options :

6406532474541. ✖ TRUE

6406532474542. ✔ FALSE

is label
it is a classification
task or not
Text gen task
BDBN

Section Id :

64065351462

Section Number :

13